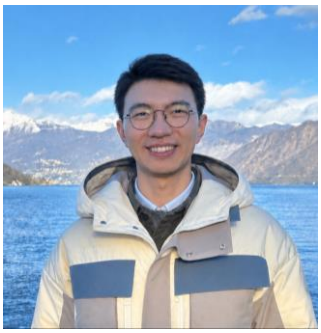
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Education

University of Jyväskylä, Department of Biological and Environmental Science — *Jyväskylä, Finland*
Joint PhD Researcher, Supervisor: Prof. Otso Ovaskainen — *2025–Now*
Beijing Normal University, College of Life Sciences — *Beijing, China*
PhD student, Ecology, Supervisor*: Prof. Anru Lou* — *2023-Now*
China Agricultural University, College of Resources and Environmental Sciences — *Beijing, China*
MSc, Resource utilization and plant protection, Supervisor: Prof. Xiaoming Shao — *2020-2023*
Northeast Agricultural University, **College of Resources and Environmental Sciences** — *Harbin, China*
BSc, Ecology, Supervisor: Prof. Shoaling Zhang — *2016-2020*

Research Experience

Joint PhD Research: *Community assembly, temporal stability and eco-evolutionary modelling* (2025-now)
China Scholarship Council (CSC): (Grant No. 202506040049)

- Compare community turnover and compositional stability across birds, arthropods, soil fungi and airborne fungi using LIFEPLAN observations.
- Contribute to JSDM Studio, a reproducible workflow for analysing ecological communities across distinct joint species distribution models.
- Use HmscEE to examine how environmental responses, lineage movement and changing palaeogeography shape historical biodiversity patterns.
- Research on biodiversity diversity distribution and model construction[24-26]

PhD Research: Study on the Spatial Distribution Pattern and Adaptation Mechanism of Tibet Bryophytes (2023-now)
Natural Science Foundation of China: Study on the genetic diversity and migration and diffusion patterns of *Syntrichia caninervis* populations (Grant No. 31870511), The functional traits of bryophytes and environmental modification effects in the arid regions of Tibet (Grant No. 41771054)

- Investigate the variations in bryophyte diversity and community stability over time and across spatial scales, and explore the reasons for the formation of bryophytes diversity patterns [10,13,18,19].
- Update the catalog of bryophyte species in Tibet and establish the Tibet Bryophyte Diversity and Trait Database (TibBryDat);
- Explore from the perspectives of macroecology and macroevolution how climate and geographic diversity impact the species diversity, functional traits, and evolutionary processes, using mechanism-based biodiversity process models (GEN3SIS, JSDM) and large language models.
- The application of different taxa and biodiversity models in ecology [17,20-23].

Master Research: Diversity and Spatial Distribution Patterns of Bryophyte Community in Tibet (2020-2023)

Natural Science Foundation of China: The functional traits of bryophytes and environmental modification effects in the arid regions of Tibet (Grant No. 41771054)

- Explored the impact of environmental filtering (climate and soil factors), biotic filtering (co-occurrence patterns and cohesion), and neutral processes on the assembly of bryophyte communities at a regional scale in Tibet [7,9,15,16,19].
- Found adaptation of traits to the environment [15].
- Predicted the potential distribution of the endangered bryophyte species *Encalypta buxbaumioidea* in China under current and future climate scenarios [12].
- Based on field survey and specimen identification, a newly recorded species of bryophytes in Tibet was reported [11,14].
- Provided an overview of the use of neural networks in the field of bryophyte ecology research [6].

Bachelor Research: Spatiotemporal Distribution of Soil Nutrients (2017-2020)

Natural Science Foundation of China: Study on the Driving Mechanism of Temporal and Spatial Differentiation in Nitrogen Erosion of Black Soil Tillage Layer under Freeze-Thaw Action (Grant No. 41771313)

- Employed Pearson correlation analysis, multivariate analysis of variance, and radial basis function neural network to elucidate the primary effects of freeze-thaw cycling on nitrogen bioavailability in the cultivated layer of mollisol [5].
- Examined the vertical distribution of soil available nitrogen within a 0-30 cm profile under conditions of freezing temperatures, soil bulk density, soil moisture, and freeze-thaw cycling frequency using the incubation method [5].

Songhua River water environment and practical research (2017-2022)

- Focused on integrating various AI technologies, including artificial neural networks and genetic algorithms, for environmental monitoring and analysis in the Songhua River [1,4,8].
- Instrumental in creating the Water Quality Assessment System for typical rivers and lakes of the Songhua River using Augmented Reality technology. This system revolutionized traditional methods of water quality analysis by incorporating interactive and enhanced reality features [3,8].
- Played a key role in developing the Ecological Risk Monitoring System for the Nenjiang River Basin, based on a mobile app platform. This system enabled real-time monitoring and risk assessment of ecological changes [2].

Teaching and Public Speaking Experience.

- **Advanced Experimental Design PhD Course Instructor** at Northeast Agricultural University, 2023: Instructed PhD students on how to design experiments and conduct field sampling, teaching various ecological data analysis methods including ordination analysis, structural equation modeling, species distribution models, and mixed-effects models.
- **Delivered the presentation "Application of ChatGPT in Ecological Research"** at Nanjing Forestry University and Ningxia University, which was one of the most attended ecology online lecture in China, attended by over 8,000 participants and streamed to 18,000 viewers, co-hosted by the "Journal of Plant Ecology" and the "Biodiversity Science". The related article was published in *Biodiversity Science*. Six days after its release, "Application of Large Language Models in Biodiversity Research" became the most downloaded article among those published in the journal this year [17].

Serving as a reviewer for "Conservation Biology" (2023) and "Methods in Ecology and Evolution (2024)".

- Review the accuracy and effectiveness of analytical methods, such as joint species distribution models.
- Determine the originality, scientific integrity, methodological correctness, and rationality of the conclusions of the paper.

Research skills

- Proficient in data analysis, demonstrating adaptability in applying techniques such as structural equation modeling, species distribution modeling (including Stacked Species Distribution Model and Joint Species Distribution Model), and utilizing tools like the GEN3SIS ecological evolutionary engine. Additionally, have expertise in ecological multivariate analyses (CCA, RDA, DCA, PCoA, NMDS, GLM) ;
- Familiar with Artificial Intelligence Algorithms, such as Artificial Neural Networks, Random Forests, and Convolutional Neural Networks; proficiency in using a wide range of analytical software including R, GIS, Google Earth Engine, Gephi, Origin, and SPSS.
- Have a solid foundation in measuring basic morphological traits for bryophyte classification. Possess excellent skills in microscopic photography and field photography techniques.
- Proficient in conducting measurements of essential physicochemical properties of soil.
- Engaged in the classification and ecological research of bryophytes, possessing extensive field sampling experience and adept in using various field measurement tools. Profoundly knowledgeable and practically skilled in botany and community ecology.
- Translation skills are solid, having translated 3 books “Joint Species Distribution Modelling: With Applications in R”, “Metacommunity Ecology” (forthcoming).

Public Engagement

“Ecology document sharing” WeChat official account resident editor, “Community Ecology” WeChat official account founder.

- It is committed to popularizing, interpreting and studying the “100 influential ecology papers” (British Ecological Society), the “Centennial Special Feature in Ecology” (American Ecological Society), and the “100 articles every ecologist should read” (Nature Ecology & Evolution) in China. Related interpreting articles were posted in WeChat official account. Received approximately 160,000 readings.
- Tutorials and videos on research methods have been published on the official account, such as “Writing Science” course, “Joint species Distribution Model for Joint Modeling of Community Ecology”, “GEN3SIS principles and R practices for studying Biodiversity patterns based on Ecology and Evolutionary processes” and so on. Received approximately 96,000 readings.

Awards

- First Prize for Outstanding Graduate Student Presentation at the Annual Conference of Chinese Bryology.
- Silver Award at the Enactus Science and Innovation China Championship.
- Second Prize for Graduate Academic Presentation at the 1st Academic Discussion of the Chinese Bryophyte Congress.
- Silver Award at the 5th Heilongjiang Province “Internet Plus” Competition.
- Gold Award at the Northeast Region of the Enactus Social Science Innovation Competition.
- Third Prize for Academic and Technological Work, Beijing Municipal “Challenge Cup” Award.
- First Prize for Academic and Technological Work, “Farming Cup” Award, China Agricultural University.
- First-class Academic Scholarship, China Agricultural University.
- Second Prize for Graduate Forum Academic Presentation, Department of Ecology, China Agricultural University.
- Second Prize in Choir Competition at the Chinese Student Art Exhibition and Performance.

Publications

2026

[26]. Zhang, Z., Bede-Fazekas, Á., García Molinos, J., Mammola, S., Kass, J. M., Qu, J., Oeser, J., Yuan, S., Zhang, C., **Gu, J.**, Ding, L., & Lin, Q. (2026). Integrating expert range maps and opportunistic occurrence records of marine fish species in range estimates. *Conservation Biology*, 40(2), e70154.

[25]. Liu, Y., Liu, S., Bede-Fazekas, Á., Mammola, S., Ding, L., **Gu, J.**, Nakamura, G., Lin, Q., Wang, D., & Zhang, Z. (2026). Factors influencing species distribution model performance in tropical reef fishes. *Diversity and Distributions*, 32(2), e70160.

[24]. **Gu, J.**, Lai, J., Wang, Y., Wu, H., Zhang, X., Song, X., Shao, X., & Lou, A. (2026). Theoretical foundations, methodological advances, and applications of joint species distribution models with a focus on the HMSC framework in ecology. *Biodiversity Science*, 34(1), 25364.

2025

[23]. Song, X., Hong, J., **Gu, J.**, Shao, X., & Wang, R. (2025). Two newly recorded species of Anthocerotophyta from Xizang, China. *Xibei Zhiwu Xuebao*, **45**(1), 158–162.

[22]. Zhang, Z., Kass, J. M., Bede-Fazekas, Á., Mammola, S., Qu, J., Molinos, J. G., **Gu, J.**, ... & Lin, Q. (2025). Differences in predictions of marine species distribution models based on expert maps and opportunistic occurrences. *Conservation Biology*, e70015.

2024

[21]. Tu, W. Q., Lu, W. X., **Gu, J. Q.**, & Lou, A. R. (2024). The species diversity and phylogenetic structure patterns of desert plant communities in the Turpan-Hami region, Xinjiang. *Global Ecology and Conservation*, *55*, e03239.

[20]. Huang, H., Zhang, Z., Bede-Fazekas, Á., Mammola, S., **Gu, J.**, Zhou, J., ... & Lin, Q. (2024). Cross-validation matters in species distribution models: a case study with goatfish species. *Ecography*, *2025*(1), e07354.

[19]. **Gu, J.**, Song, X., Ye, Y., Shao, X., Liao, Y., & Shao, X. (2024). Exploring the driving factors of bryophyte assemblage distribution patterns in Tibet. *Frontiers in Ecology and Evolution*, *12*, 1376263.

[18]. Zheng, G[#]., **Gu, J.**[#], Zhao, W., Zhang, Y., Guan, Z., Lei, M., & He, C. (2024). Spatial, Geographical, Climatic, and Edaphic Influences on Moss Community Structure: A Case Study from Qinhuangdao, China. *Forests*, *15*(3), 424. ([#]These authors contributed equally as co-corresponding authors)

[17]. **Gu, J.**, Chen, J., & Lai, J. (2024). Application of large language models in biodiversity research. *Biodiversity Science*, 24258.

[16]. Song, X., **Gu, J.**, Ye, Y., Wang, M., Wang, R., Ma, H., & Shao, X. (2024). Exploring Intraspecific Trait Variation in a Xerophytic Moss Species *Indusiella thianschanica* (Ptychomitriaceae) across Environmental Gradients on the Tibetan Plateau. *Plants*, *13*(7), 921.

[15]. Song, X., **Gu, J.**, Liu, L., Liao, Y., Ma, H., Wang, R., ... & Shao, X. (2024). Exploring the distribution and habitat preferences of Polytrichaceae (Bryophyta) in Tibet, China. *Heliyon*, *10*(14).

[14]. Sun, Y., Song, X., Chen, C., Li, S., **Gu, J.**, & Shao, X. (2024). A New Record of *Pogonatum tahitense* (Polytrichaceae) from Tibet, China: Taxonomic Description, Range Expansion, and Biogeographic History. *Plants*, *13*(6), 846.

2023

[13]. Song, X.[†], **Gu, J.**[†], Ye, Y., Li, W., Liao, Y., Wang, R., Ma, H., & Shao, X. (2023). The Diversity and Community Pattern of Liverworts on Sygera Mountain, Tibet. *Forests*, 15(1), 48. ([†]These authors contributed equally to this work)

[12]. Liao, Y., Song, X., Ye, Y., **Gu, J**., Wang, R., Z., Zhao, D., & Shao, X. (2023). Climate Change May Pose Additional Threats to the Endangered Endemic Species *Encalypta buxbaumioidea* in China. *Diversity*, 15(2), 269.

[11]. Hong, J., Song, X., **Gu, J.**, Li, W., Dong, Y., Zhuo, G., & Shao, X. (2023). Calyculariaceae, a newly recorded family of bryophytes in Tibet, China. *Journal of Plant Resources and Environment*, 5, 95-97.

2022

[10]. **Gu, J.**, Song, X., Liao, Y., Ye, Y., Wang, R., Ma, H., & Shao, X. (2022). Tree Species Drive the Diversity of Epiphytic Bryophytes in the Alpine Forest Ecosystem: A Case Study in Tibet. *Forests*, 13(12), 2154.

[9]. **Gu, J.**, Song, X., Liao, Y., Ye, Y., Wang, R., Ma, H., & Shao, X. (2022). *Elevation, climate, and tree property drive the diversity of epiphytic bryophytes in the alpine forest ecosystem, Tibet*. The International Association for Ecology Conference, Geneva, Switzerland (Poster presented at the International Association for Ecology Conference).

[8]. **Gu, J**. (2022). Application of Augmented Reality in the Field of Water Environment. Liu, B., Xie, G., Cao, G., Zheng, G. (eds.), *Water Environment Assessment and Intelligent Prediction in Typical Areas of Songliao Basin*. Beijing: Science Press.

2021

[7]. **Gu, J.**, Song, X., Liao, Y., Jiang, X., & Shao, X. (2021). *Species diversity and spatial distribution pattern of epiphytic bryophytes in alpine forest ecosystem of Sejila Mountain*. The 9th East Asian Federation of Ecological Societies International Congress, Hohhot, China (Oral Report and Conference Abstract).

2017-2020

[6]. **Gu, J.**, Liu, L., Liao, Y., Song, P., & Shao, X. (2020). *Application of Neural Network in Bryophyte Ecology*. The 1st Chinese Bryophyte Congress (pp. 9). (Oral Report and Conference Abstract)

[5]. **Gu, J.**, Tong, Z., Wei, Y., Liu, X., Feng, W., Zhu, X., & Zhang, S. (2020). Dynamics of Soil Available Nitrogen Influenced by Freeze-Thaw Cycling in the Profiles of Plough Layer in Mollisol. *Journal of Soil and Water Conservation, 35*(2), 295-302.

[4]. **Gu, J.** Application of Artificial Intelligence Technology. (2020). In: Gao, F., Liu, W., Zheng, G. (eds.), *Water Environment Monitoring and Evaluation of Provincial Buffer Zone in Songhua River Basin*. Beijing: Science Press.

[3]. Liu, B., Ma, X., Zheng, G., Cao, G., **Gu, J.**, Yu, J., & Xie, G. (2020). Water Quality Assessment System for Typical Rivers and Lakes of the Songhua River Using Augmented Reality (AR) Technology (Computer Software Copyright No. 2020SR1185814).

[2]. Zhang, D., Zheng, G., Zhao, W., Jia, X., Zhao, Y., & **Gu, J**. (2018). Ecological Risk Monitoring System for the Nenjiang River Basin Based on Mobile App (Computer Software Copyright No. 2018SR108640).

[1]. Zheng, G., Wang, Z., He, J., & **Gu, J**. (2017). Water quality assessment of the Songhua River provincial boundary buffer zone based on matter-element analysis [J]. *Northeast Water Resources & Hydropower,* 35(12), 25-27.